TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

Reminders

Distribute first: multiply the outside number by every term inside.

Get x alone: undo + and − first, then × and ÷. Do the same thing to both sides.

Check: put your answer back in. Both sides equal → correct.

How many solutions? Same x-term and same number: infinite. Same x-term, different number: none. Otherwise: one.

1. What value of p makes the equation $7 - 3(p + 2) = 10 - (2p + 5)$ true?

Tip: Distribute on both sides first. −( ) changes every sign inside.

A. 4

B. −6

C. **−4**

$1 - 3p = 5 - 2p \rightarrow -p = 4 \rightarrow p = -4$

2. Solve the equation shown.

$0.5(4x - 6) = -1.5x + 4$

> **Tip:** Distribute the 0.5 to both terms.

Distribute: _____ = −1.5x + 4

Add 1.5x to both sides: _____

Add 3 to both sides: _____

Divide: x = _____

$x = 2$

$2x - 3 = -1.5x + 4 \rightarrow 3.5x = 7 \rightarrow x = 2$

3. Does the equation shown below have one solution, no solution, or an infinite number of solutions? Explain your reasoning.

$\frac{1}{2}(8x + 6) = 2(2x + 1) + 2$

> **Tip:** Simplify each side, then compare the x-terms and numbers.

Left side simplifies to: _____

Right side simplifies to: _____

Circle: one solution · no solution · infinite

No solution

$4x + 3 = 4x + 4$: same x-coefficient, different constants, so $3 = 4$ is never true